FINTECHATHON 2026 · TOPIC E: SME FINANCE COPILOT

DecisionGuard

A finance copilot for small Bangladeshi shops. Code computes every number; the model only explains it, and the owner approves every action.

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Small firms lack credit, and the records to get it

$5.7 trillion

finance gap for micro, small and medium firms in developing economies (IFC)

$2.8 billion

credit gap for Bangladesh’s small businesses (IFC estimate)

7.28%

of Bangladesh’s small-business credit goes to firms led by women

71%

of 9,252 Chinese small firms surveyed were owed unpaid receivables

Sources: IFC MSME finance gap factsheet; IFC via Softograph; Dhaka Tribune; Peking University survey reported by SCMP

An owner with three questions

Her records today

- bKash and Nagad statements
- a bank account and a cash book
- a paper *baki* register of credit sales
- informal loans when cash runs short

A · FINANCING

“Which loan is really the cheapest, and can I get it?”

B · CASH FLOW

“Will I run short of cash, when, and what can I do now?”

C · ANALYSIS

“Is my business actually doing well, and what if I decide this?”

Every shop in our demo is synthetic and labelled so: five invented Dhaka businesses, 18 months of daily records

WHY NOT JUST A CHATBOT?

Code computes.
The model narrates.

Language models are not calculators. They pick the wrong figure, rarely catch their own mistakes and sound surer than they are. With money, a guessed number is a wrong decision.

Research summary: technical documentation §1

How DecisionGuard works

Shop records

bKash, Nagad and bank statements, cash book, baki register, receipt photos

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Typed facts

Bangla and Banglish read; names and phone numbers turned into tokens; every row keeps its source

→

Engines

Cash, financing, analysis, netting: exact decimals, seeded simulations, a formula for every figure

→

Copilot

Plans and explains; every number it states is recomputed before it is shown

→

Owner

Approves anything that leaves DecisionGuard, with a one-time code

Tap any number on screen and DecisionGuard shows its formula, its inputs and the source rows behind it.

B · Warns before cash runs short

13 weeks ahead, 1,000 simulated paths built from the shop’s own weekday, month-end and Eid patterns and each customer’s payment habits.

Alerts in three tiers (Watch, Warning, Critical), repeated only when the risk grows.

Every fix is measured on the same simulated paths, then offered as an action card the owner can approve.

Synthetic hardware shop, as of 30 Sep 2026

86.7%

chance of a cash gap that starts in 5 days; expected shortfall Tk 73,460.90

Cheapest fix: ask a buyer to pay early. It closes Tk 71,736.88 of the gap at no cost.

Backtest on the five synthetic shops: the one-week forecast error beats a “no change” forecast at all five (technical documentation §3.3)

A · Ranks loans by what they really cost

HEADLINE RATE 9%

11.29%

true yearly cost: bank loan, reducing balance, 0.5% fee

HEADLINE RATE 9%

22.15%

true yearly cost: digital loan repaid in daily debits, 0.575% fee

13 real Bangladeshi products, each with its source link, a quoted line and the date we checked it

A fact we don’t hold shows as “not on record”, never as a pass

“2% off if you pay 10 days early” costs 109% a year

Tk 50,000 for 180 days; rates and fees from the providers’ pages, checked 18 Oct 2026 (data/public/financing/bd.yaml)

C · Compares decisions before you make them

Decision (synthetic hardware shop)	Chance below the safety buffer	Cost
Do nothing	93.9%	Tk 0
Delay a week of purchases	93.8%	Tk 0
Take the best loan (pays out 30 Oct, too late)	92.4%	Tk 3,224.41
Net debts with other shops	83.1%	Tk 0
Offer 2% off for paying early: recommended	**51.9%**	**Tk 7,600.00**

Stability 0.80: the advice held in 8 of 10 re-runs. It changes if sales fall 20% or a surprise bill arrives, and never when rows are shuffled, customers renamed or the question reworded.

1,000 simulated paths in 40 economic states (sales, prices, late payers, rates, shocks); cost = fall in expected cash at 13 weeks

A copilot that checks itself, and says “I don’t know”

1 Guard

Screened for prompt injection in four languages

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2 Plan

A model proposes steps; code drops anything unknown

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3 Check

Every number recomputed independently; mismatches removed

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4 Challenge

Another model family argues against; others vote

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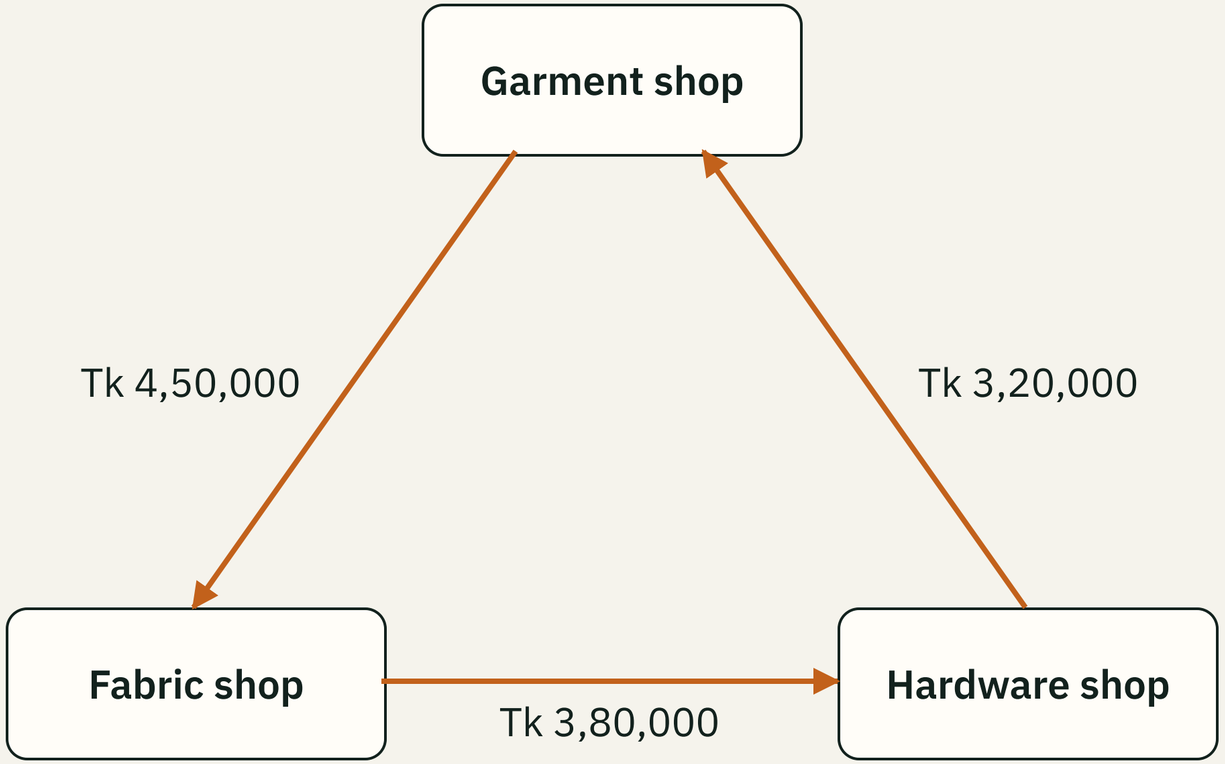
5 Answer or abstain

Withheld when checks fail, naming what is missing

“How big a loan can I safely take?” 14 of 14 numbers recomputed and verified, 3 action cards, and a warning that the lender may ask for facts DecisionGuard does not hold.

Run offline with no model, so no model agreement is claimed. Each question has a budget of tool and model calls; agent tools can never execute an action.

Clears circular debt between shops with less cash



Tk 11,50,000

of debt cleared with

Tk 1,90,000

of cash, along debts that already exist

Every owner approves with a one-time code. The settled record is sealed with a salted SHA-256 hash for Ethereum Sepolia: only the hash, never names or amounts.

Synthetic ring; arrows run from the shop that owes to the shop that is owed. In our evidence the anchor is offline.

Money never moves without the owner

Tier	Who and how
READ	Look at figures and their sources
SUGGEST	Propose an idea to someone with more rights
DRAFT	Prepare something that stays inside DecisionGuard
EXECUTE	Anything that leaves DecisionGuard: the owner approves with a fresh one-time code, within 24 hours, and the policy is checked again when it runs
DENY	Nobody, not even the owner: change bank details, add a payee, switch off the audit log, export raw personal data

Audit log chained by hashes; the database refuses edits; the chain’s head is anchored every night

Privacy: only anonymised text can reach a model, and the gate fails closed

Injection guard in four languages: 96 of 96 known attacks, 0 of 96 false alarms. On 12 unseen attacks it first caught 3, yet the copilot acted on none; widened rules now catch all 12, still with no false alarms.

Full tier table, threat model, OWASP Top 10 for LLM applications and 21 known risks: security self-assessment

Built, tested and logged

2,450+
automated tests; none calls a model or the network

48
steps in our sandbox log through the real API, all passing

13
loan products, each with a source link, a quote and a check date

3
languages in the app: English, Bangla and Chinese

Every pull request runs lint, tests, dependency audits and a secret scan, then boots the full Docker stack and checks TLS and security headers.

Evidence index: docs/submission/evidence in the repository

What is not proven yet, and what comes next

Honest limits

- All demo data is synthetic; no real shop has used DecisionGuard yet
- The injection guard is rules only; after widening, it still needs a fresh unseen test
- The copilot was measured offline, not with live models at scale
- Reduced: receivables and message drafts (no sending); cut: market watch

Next

- Pilot with consenting Dhaka shops, from their own bKash and Nagad exports
- Sending approved reminders through bKash and SMS partners
- Licensed lenders to fund netting and confirm track records
- A learned injection classifier beside the rules, and an external security review

Numbers you can check.
Actions you approve.

Thank you ধন্যবাদ 谢谢

github.com/NoboIshraque/SME-financing-copilot

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